Opening

Title

- Hoapa = class of communication systems

- optimized for collective inquiry
- generalization seems relevant in Nature

- name: last slide

My name

- as good as their theories (footballer-feet)
- science $\cong$ racing
- Univ dev $\cong$ more and more demanding cars

- I built my own car:

- by chance (brain soup)
- as an exercise/specs

- "what is… U&B" $\rightarrow$ I trusted it $\rightarrow$ "took for a ride"

- 1st step, labels: "the system doesn't know…"
- last: "stays in the abstract, as available resource, no movement back into the concrete"
- vs. loss of knowledge, useless
- complicated, non-trivial thing $\rightarrow$ workshops

- logic: odd workshops $\rightarrow$ special workshops

Pitch

[…]

- not a pitch but one of them
- inherently perspectivist
- a sort of motive

Permacultural

- abstraction of the agricultural technique
- targets minimal imprint (your system ≅ the natural's)
- requires knowledge (your knowledge ≅ the system's)

Scientific

- an extension not a modification/relaxation
- a protocol becomes an object of science
- scientists (anyone) expected to sign up for the most trustworthy
- trust := whatt guarantees […]
- "layers of complexity"

Inventory diagram

- layers ≅
- separations of concerns ≅ specs
- poles of adjunction ≅ motivation &

Sources

A. Grothendieck

- workshops (embryonic)
- $\cong$ col. sci. method (specific use case)
- $\cong$ recursive elem. in G-evol (in Nature)
- main reason to believe in permacultural logic?

Albert Lautman

- bio (1930, anticipator, non-reductionist)
- more recent anchor

- demystifying metaphysics
- movement from metaphysics to effective theories
- giving a context to mathematics, physics… Hoapa

Scientists

- each step = a correspondance test

Thesis

Work

- engine → Peter Atkins (extending physics)

- work = "constrained release of energy into a few degrees of freedom"

- organism → Kauffman (origins of life)

- closures = "collective work which constructs and maintains the constraints that enable those processes" (constraint, work-task, catalytic)

- computer (also brain, genome, communities) → Pattee (symbolic and dynamical)

- "physical activity is recorded in symbols, which can be stored, manipulated and interpreted to guide further physical activity" (von Neumann, Peirce)

- logic → Lautman: formal system and realization (for us: abstract/concrete)

- Hoapa = a system of constraints, open but capable of closure, and optimized for symbolic work

Dedicated

- evident in computers & genome:
- ""

- brains are less clear-cut (b/c internal concrete/abstract divides, as in math and code)

- communities are "degenerate"
- concrete substrate not just coupled, dependent on the abstract

Decomposition

- dim = decompose through the structure of the ambient space
- prop = decompose through the structure of the element itself

- at the community level → knowledge
- brains are like vectors
- documented statements like prime factorization

Theory of work

- "Constraints give you effects, closure gives you continuity, the abstract gives you extra possibilities"

Project

Open

- its own experiment

- autonomous = managerless
- role-symmetric = all play by the same rules…
- self-reflexive = …they themselves set
- modular = well-engineered constraints
- transparent = pedagogical and integrated

- federated & network-shaped ⇒ local respectful of global
- sheaf-like ⇒ global respectful of local (unlike any theory of "social contracts")

- → UTL: open ≅ accessible

Results

- illustrates the division of labour

- crucial point where the verification touches the implementation
- adapter/translator/buffer between the "synthetic" agents and their "analytic" trace

- route to bigger adjunction

Process

Overview diagram

- convs → atoms of reflexion
- concurrency → network of reflexion
- graphs → structured memory of reflexion
- Hoapa as Jacob's bricolage

Conversation

- amazing results when experimented in politics
- random selection and consensus ↔ "play by the same rules"
- "let it crash" in Erlang

Together

- these concurrency models follow Lautman decomposition:

- actors are "synthetic" agents

- their messages "analytic" decomposition

Graph

- the "base" layer = "verbatim" orchestration
- refinement into specialized/pruned graphs
- recap: conv (born) → concurrency (travels) → graph (lands)

Independently

- not much change in content (broader scope)
- impact is on diffusion & accessibility, trust, usage & usefulness, representativity, control of new technologies…

Partial diagram

- Lautman's demystified metaphysics
- → theory of abstract work
- → implementation, verification, conjecture

Verification 8/41

Referee

- ∅

Verification

- ∅

Participants

- ∅

Conjecture 9/41

Adjunction

- see later

Emerge

- Simondon's view: context of potentials, coevolution, never exhausting it
- candidate for future individuation from new context: AGI

- reusability (return to context)
- or "standalone byproduct" (new context, replay)

Universal property

- a translation

Rising sea

- Permacultural logic is an outgrowth of my work on Hoapa, not one of its constituents
- It could be that Hoapa's optimality is true and the permacultural interpretation of logic is false
- main reason to believe is not technical, but that the rising sea methodology works

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Permacultural logic

Harmony

- Gentzen in intro/elim
- in type theory with how to construct a type and how to use a type
- Hilbert in axioms and metamathematical theory (in the vision of Lautman)
- My eureka moment came from a biological analogy: "A cell draws on a proton gradient to produce ATP for new work, while consuming fuel to replenish the gradient, some of that work maintains the machinery that keeps the cycle going"
- see later

Replay

- As an intuition:
- visible part is the feedforward: new problems leading to new results (à la cybernetics)
- "you can't see the feedback, but for the structural rules to hold (≅ "the system to be sustainable"), there must be a form of non-exhaustion and positive feedback"

Development

Unpredictable

- "only collective intelligence should shape what collective intelligence should be"
- the same form of non-ergodicity as biology

Stake

- stakes are facilitators of consensus (in citizen assemblies, high gain)
- In Hoapa, so are role-symmetry and general equalitarian conditions, openness, anonymity… (low cost)

Documentation

- Grothendieck in 1972, when asked about "the revolution":
- he first discards the top-down approach
- then calls for a bottom-up approach (start with "fixing" the relations between individuals)
- and uses the example of the teacher-student relation.

A park

Operational documentation

- *literate programming* as working model (the great missed opportunity of software development)
- better model is team semantics and provenance
- "for the work to have document-to-document continuity, as opposed to brain-to-brain or self-to-self, the documentation needs to show how a result was obtained, what it depends on, and what may need reconsidering when those dependencies change, etc."

Science

*

- "And all science, when we do not understand it as an instrument for power and domination, but as an adventure in knowledge undertaken by our species through the ages, is but this harmony, more or less wide and more or less rich from one period to the other, which unfolds over the generations and centuries, by the gentle counterpoint of all the themes emerging in turns, as if summoned from the abyss, to join and embrace in her" (Grothendieck, R&S)

- "From a semiotical and philosophical perspective, the general method is fascinating: to understand something, consider all possible points of view and search for a structural connection between them. Only then you will be approaching true knowledge. A systematic use of the liberty and multiplicity of topos theory in our ordinary life would have huge consequences on tolerance, ethics, politics, and social action: an example of how high mathematical practice would transform our everyday practice." (Zalamea, 'Grothendieck's 40 Main Years: A Unitary Vision Through the TSK Models')

Language

- From React's slogan: systematize correspondences in the vocabulary and symbols across specialized domains

- Homotopy theoretic work on the graphs of the protocol → a landscapes of concepts (plane of immanence of Deleuze)
- peaks/special points → words of a language
- give them symbols and a grammar to interact and form structures and statements (optimized for accessibility, not specialized usage) → a universal technical language
- the "rest" is preserved in the "abstract infinitesimal variations" (the "in between the words")
- it is in dialectic tension with the "concrete language substrate" → third separation: 1. concrete/abstract 2. synthetic agents/analytic documentation 3. language/variations
- technical bridge: "Awodey and Forssell's First-Order Logical Duality, which explicitly connects with the Grothendieckian setting, relating categories presenting theories to topological groupoids of models and their isomorphisms, using topos theory to establish a syntax–semantics duality."

Mathematics

Further

- Kennedy, Magidor and Väänänen - "Inner Models from Extended Logics", or Kennedy's essay "On the "Logic without Borders" Point of View" as concrete examples of what "a mathematical world generated through specified expressive resources" can mean
- main "reason to believe": rising sea methodology (explain)

Generalized permaculture

Networks

Care

- Some related structures:
- Lautman's "solidarity networks"
- Yoneda lemma
- game-theoretic semantics, particularly the tradition of Hintikka, Sandu, and Väänänen. For instance 'Abramsky and Väänänen - From IF to BI: A Tale of Dependence and Separation' ("In game semantics, negation exchanges the verifier's and falsifier's roles: a mathematical precedent for treating reciprocity of roles as part of an interaction's structure")

Informal recap diagram

- looking for a mathematical setting in which the questions motivating the design become questions we can investigate precisely

Motivations and goals

Optimality

- *understanding* = demanding abstract work

Perspectivism

- Philosophical perspectivism with its long tradition (Deleuze, Spinoza)
- Lautman anticipates sheaf-theory with global/local
- in HLG scope: Ehrenfeucht–Fraïssé games, Abramsky's contextuality work, and its translation into team semantics with Puljujärvi and Väänänen

Presuppositionlessness

- The condition for openness + dedication to the concrete/abstract work relation
- VS. every assumption coming in and staying fixed is an unnecessary constraint
- "reduces to one assumption: the assumption-free minimal open system which permits balanced community communication is also the special or optimal one (amounts to trusting communication itself, unlike the entire field of "collective intelligence")"

Categories from metaphysics

F

- Complexity in the sense of Yuri Manin

- an alternative entry point into Hoapa, like Kauffman/Simondon is

- Nature operates at a constant level of complexity
- any apparent variation reflects a change in perspective…

G

- "Darwinian evolution generalizes, but the part that generalizes is not selection, but recursion: what evolution produces becomes a condition for what can evolve next"

η

- It's a mistake to remove the process of inquiry from the object of the inquiry

- "What is mathematics, really?" by Ruben Hersh

ε

- Nature operates at a constant level of complexity

- any apparent reduction reflects what is forgotten by the functors mediating between the world and an observer's perspective

Read aloud

On the slide	Say
$F : Ctx \to Inq$	"F, from context to inquiry"
$G : Inq \to Ctx$	"G, from inquiry to context"
$\mathrm{Hom}_{Inq}(F\ c, i) \cong \mathrm{Hom}_{Ctx}(c, G\ i)$	"Hom in inquiry, from F of c to i, is in natural bijection with Hom in context, from c to G of i"
$\eta_c : c \to GF\ c$	"eta c, from c to G F of c"
$\varepsilon_i : FG\ i \to i$	"epsilon i, from F G of i to i"
(L, C*, HOD)	"L, C star, H-O-D"
$w : F\ c \to i$	"w, from F of c to i"
$G(w) \circ \eta_c : c \to G\ i$	"G of w after eta c, from c to G of i"

Permacultural logic

- AI generated, motivated by all of the above

Read aloud: from work to reusable construction

On the slide	Say
$(- \times A) \dashv (A \Rightarrow -)$	"dash times A is left adjoint to A implies dash"
$\mathrm{Hom}(\Gamma \times A, B) \cong \mathrm{Hom}(\Gamma, A \Rightarrow B)$	"Hom from gamma times A to B is in natural bijection with Hom from gamma to A implies B"
$w : \Gamma \times A \to B$	"w, from gamma times A to B"
$w\sharp : \Gamma \to A \Rightarrow B$	"w sharp, from gamma to A implies B"
$(A \Rightarrow B) \times A \to B$	"A implies B, times A, to B"
β and η	"beta and eta" (here the eta law, not the unit)

Read aloud: what does the work leave available?

On the slide	Say
$\mathrm{State}_A(X) = A \Rightarrow (X \times A)$	"State A of X is A implies X times A"
$\mathrm{Store}_A(X) = (A \Rightarrow X) \times A$	"Store A of X is A implies X, times A"
$p(a) = (x, a_0)$	"p of a is the pair x, a nought"
$q' \circ w = \bar{w} \circ q$	"q prime after w equals w bar after q"

Reading guide (from *Categories from metaphysics* on)

Categories and arrows

- Ctx — say "context" (the category of contexts); never spell out C-T-X
- Inq — say "inquiry" (the category of organizations of inquiry); never spell out I-N-Q
- F : Ctx → Inq — "F, from context to inquiry" (or "F, a functor from Ctx to Inq")
- F c, G i, GF c, FG i — "F of c", "G of i", "G F of c", "F G of i"
- Hom_Inq(F c, i) — "Hom in Inq, from F c to i"; Hom rhymes with "Tom", not "home"
- $\cong$ — "is in natural bijection with" (or "is isomorphic to")
- $\dashv$ — "is left adjoint to"
- $\circ$ — "after" (or "composed with"): G(w) $\circ$ η_c is "G of w after eta c"
- w$\sharp$ — "w sharp", the transpose of w
- $(- \times A)$ — "dash times A", the product with A; the dash is an empty slot, not a minus
- $A \Rightarrow B$ — "A implies B" (also "B to the A", the exponential)

Greek letters

- η — eta: "EE-tuh" (or "AY-tuh"); the unit
- ε — epsilon: "EP-sih-lon"; the counit
- β — beta: "BAY-tuh" (or "BEE-tuh")
- Γ — capital gamma: "GAM-uh"; the remaining assumptions
- Careful: η means two different things. On the Categories slides it is the unit; in the Harmony line it is the η-law, so say "the eta law" there.

Subscripts, primes and bars

- η_c, ε_i — "eta c", "epsilon i" (or "eta sub c", "epsilon sub i")
- a_0 — "a nought" (or "a zero")
- q' — "q prime"
- $\bar{w}$ — "w bar"
- State_A(X), Store_A(X) — "State A of X", "Store A of X"

- $p(a) = (x, a_0)$ — "p of a is the pair x, a nought"
- $q' \circ w = \bar{w} \circ q$ — "q prime after w equals w bar after q"

Inner models (the Evolution line)

- L — "L": Gödel's constructible universe, built with first-order definability
- C* — "C star": the inner model built with the cofinality quantifier (Kennedy, Magidor, Väänänen)
- HOD — letter by letter, "H-O-D": the hereditarily ordinal definable sets, built with second-order definability

Words

- functor — "FUNK-ter"
- monad, comonad — "MON-ad" (or "MOH-nad"), "co-MON-ad"
- counit — "CO-you-nit"
- topos — "TOP-oss" (or "TOE-poss"); plural toposes or topoi
- univalence — "you-nih-VAY-lence"
- currying, uncurrying — "KUR-ee-ing", after Haskell Curry
- cartesian closed — "car-TEE-zhun closed"
- intuitionistic — "in-tew-ISH-uh-nis-tik"
- HoTT — said "hot": homotopy type theory
- elementary bound — if asked: bounded by a tower of exponentials of fixed height
- Coq — said like the French "coq"; renamed Rocq ("rock") in 2025, so "Rocq, formerly Coq" avoids the English pun
- ring — the tactic's name, said as the word

Names

- Ehrenfeucht — "AIR-en-foyt"
- Fraïssé — "fra-ee-SAY"
- Gentzen — "GENT-sen"
- Zalamea — "sa-la-MAY-a"
- Kauffman — "KAWF-mun"
- Peirce (oral commentary) — "purse", never "pierce"

- Lawvere (oral commentary) — "law-VEER"
- Grothendieck (oral commentary) — "GROH-ten-deek"
- Schrödinger (oral commentary) — "SHRUR-ding-er"

Introductory symbols

- : as in, meaning, which we take to be
- $\rightarrow$ whose function is
- $\Rightarrow$ implying, which yields, from MW we get

What the university gets wrong (Science, list of evils)

- objectivity/view from nowhere
- the conversion of expertise into authority over other participants;
- the ranking of people through their publications, affiliations, and institutional positions;
- the replacement of curiosity by the pursuit of credentials, funding, and visibility;
- the enclosure of knowledge behind paywalls, proprietary tools, and restricted access;
- the neglect of explanation, translation, replication, and maintenance because they bring less recognition than novelty;
- the dependence of entire projects on a few people whose departure makes the accumulated work difficult to continue;
- the loss of failed attempts, unresolved disagreements, and the conditions under which a result was obtained;
- the duplication of work across communities unable to find, understand, or reuse one another's achievements;
- the pressure to fit new questions into established categories rather than revise those categories;
- the treatment of specialized roles as fixed intellectual identities;
- the concentration of decisions about worthwhile questions in institutions distant from those concerned by them;
- the accumulation of results without corresponding improvements in their accessibility or usefulness;
- the neglect of the human and ecological conditions that make continued inquiry possible;
- the exemption of the organization of science itself from the scrutiny it applies to its objects.

Oral commentary from the talk text (Permacultural logic slides)

Return

Peirce describes diagrammatic reasoning as constructing a diagram, experimenting on it, and establishing which consequences hold for any construction made under the same conditions. A particular act of understanding yields a general relation. This is the movement that interests me in return: work acquires a form that can outlive the occasion of its discovery. Hoapa asks a conversation to leave such an achievement, with the relations it establishes and the conditions under which they hold available to others. See Peirce on diagrammatic reasoning, NEM 4:47–48.

Replay

Grothendieck's relative point of view gives this portability a discipline: keep the setting and the relationships through which a construction can pass to another setting. What must be established is which passages preserve what we want to use. Hoapa takes this as a design demand. Another community should be able to understand what it is importing, examine its applicability, and develop it without needing the original producers to return or authorize each use. An achievement becomes common means while its dependencies remain visible. This is our application of the relative viewpoint.

Harmony

Lawvere shows how logical operations can be characterized through their relations to other operations. That gives Hoapa a way to make a philosophical demand exact: what must using an achievement preserve of the work that made it warranted? Establishing and using have different responsibilities, but their rules must answer to one another. Here the separation/integration mirror reaches inside logic itself. The harmony between receiving and giving back acquires a precise instance, which can then guide our search for adequate relations between inquiry and context. See Lawvere's "Adjointness in Foundations," §3.

State monad

Simondon asks us to understand an individual through the process that constitutes it together with its milieu. The achieved form partially resolves a situation containing tensions and potentials; further individuation remains a separate possibility. I apply this question to Hoapa's conversations and their achievements: what has been resolved, and what situation has the resolution left for others? Solving a problem can rightly remove a tension. The permacultural demand concerns whether the capacity for further

development survives that success. See Simondon, "The Position of the Problem of Ontogenesis," pp. 5–8.

Store comonad

Kauffman's discussion of constraint closure draws attention to what an activity relies on: living processes help produce the constraints that channel those processes. This is the enabling side of work. A Hoapa document similarly relies on a maintained language, working tools, and people able to interpret it. A local task can use those conditions without reproducing their whole history, because other activities support them. That is the philosophical question I bring to the comonadic side: how does a local use remain connected to the organization that makes it possible? See Kauffman's "Answering Schrödinger's 'What Is Life?'," §16.

Continuation

Peirce's pragmatic maxim clarifies a concept through its conceivable practical consequences. This directs us toward what a community can understand and do through an expression, including uses beyond the immediate task. A publication should let later inquirers recognize the operations it supports and the points where their question exceeds it. Hoapa's universal technical language has this ambition: make relations usable across different inquiries, while allowing a new problem to demand a new distinction. This is the philosophical purpose of testing continuation. See Peirce's "How to Make Our Ideas Clear," §II.

Cut

Lautman distinguishes the constituents proper to an object from the common scheme in which objects are expressed. Hoapa uses this distinction to describe a change of role: the differentiated product of one inquiry can become part of the common language of another. A lemma remains a particular achievement while becoming an available step in someone else's work. This is what I mean when I say that the analytic aspires to become synthetic. The community builds a body of knowledge that can become a shared basis for further thought. See our reading of Lautman's proper and dimensional decompositions.

Feasibility

Peirce explicitly places an economy of research within his account of logical method. He asks how acquiring and spreading knowledge relate to the effort they require. That is close to Hoapa's motivation: the organization of inquiry determines how much earlier work can help later work. The proof-length result gives one exact instance of this concern. A community's finite capacity can be spent rebuilding what it already knows, or extending

what its achievements now let it do. Making knowledge accessible and reusable therefore participates in the development of knowledge itself. See Peirce's Carnegie application, memoirs 27–28.

Reflection

In Lautman's universal-covering example, an achieved structure makes further mathematical constructions possible. Zalamea reads such passages through saturation and liberation: enriching a setting can release new forms of construction. Hoapa asks for a similar passage in collective inquiry. Reflection makes the point tangible: a proved result becomes machinery for proving. The achievement changes its role in the activity and helps constitute the means through which subsequent achievements are produced. This is the structure/genesis relation we want to investigate. Sources: Lautman, Mathematics, Ideas and the Physical Real, pp. 130–131; Zalamea's introduction to that volume, pp. xxiv–xxv.

Renewal

Simondon calls transduction a process in which an already formed structure supports further structuring. Grothendieck's rising-sea method gives us a mathematical kinship: develop the surrounding concepts until constructions become available within them. Hoapa aims to give this movement an organized collective form. Documents become shared concepts; those concepts change the questions a community can formulate; those questions set further work in motion. This is renewal when the new setting supports a capability its predecessor did not provide. See Simondon, p. 11, and Grothendieck's Récoltes et Semailles, English translation, PDF p. 93.

Conditions of activity

Kauffman's account of biological work suggests a stronger requirement than accumulating products: the productive organization has to keep making its own activity possible. Hoapa brings this question into collective inquiry. Knowledge should help sustain people's capacity to investigate, the means they share, and the world on which both depend. This is how I understand its possible participation in a generalized evolution: a context enables inquiry; inquiry produces achievements that become part of a changed context; that context supports further inquiry, with new means and new problems. The conjecture seeks a mathematical account of this passage, and permacultural logic names the demand that development sustain what enables it.